

JMORF — Morpho-Syntax

Sign-Based Construction Grammar

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Lecture 16

Location: SV 2.39

JMORF (2025)

Overview

- Chapter 16 framework (same analyses, different underlying system)

Construction Grammar (CxG)

- A family of grammars based on the idea that
 - Knowledge of a language comes from **form and function pairings**
 - **function** includes meaning, content, or intent (both semantics and pragmatics)
 - **form** includes phonology, syntax, orthography
- CxG grew out of generative semantics and cognitive linguistics, by researchers such as Charles Fillmore, Paul Kay and George Lakoff
- Instead of language as a grammar+lexicon, think of it as a structured network of families of constructions

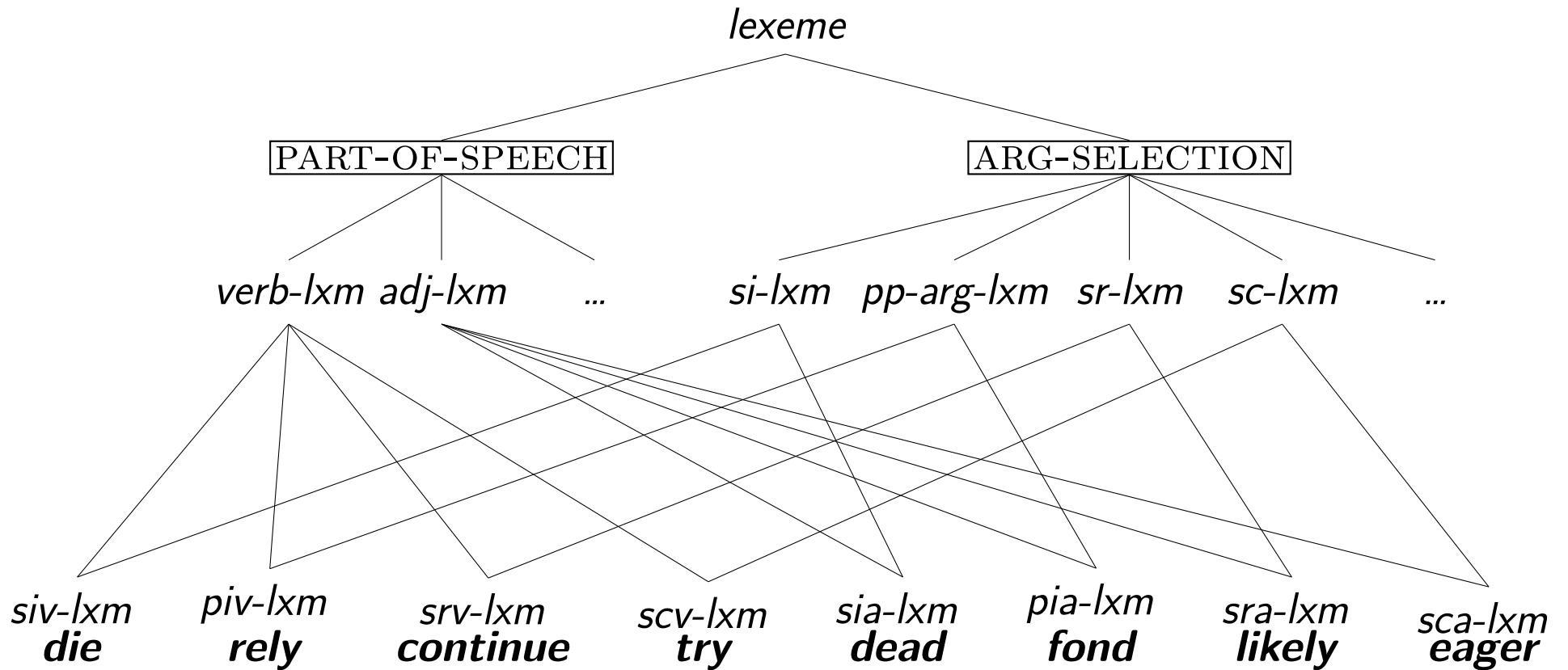
Construction Grammars

- **Sign-based-Construction Grammar** (Berkeley Construction Grammar)
unification-based framework (with computational implementation)
- **Goldbergian/Lakovian Construction Grammar**
psychologically plausible
- **Radical Construction Grammar**
syntactic categories, roles, and relations are not universal:
they are not only language-specific, but also construction specific
- **Embodied Construction Grammar**
relates constructions to embodiment and sensorimotor experience
- **Fluid Construction Grammar**
learns grammars from the environment (with computational implementation)

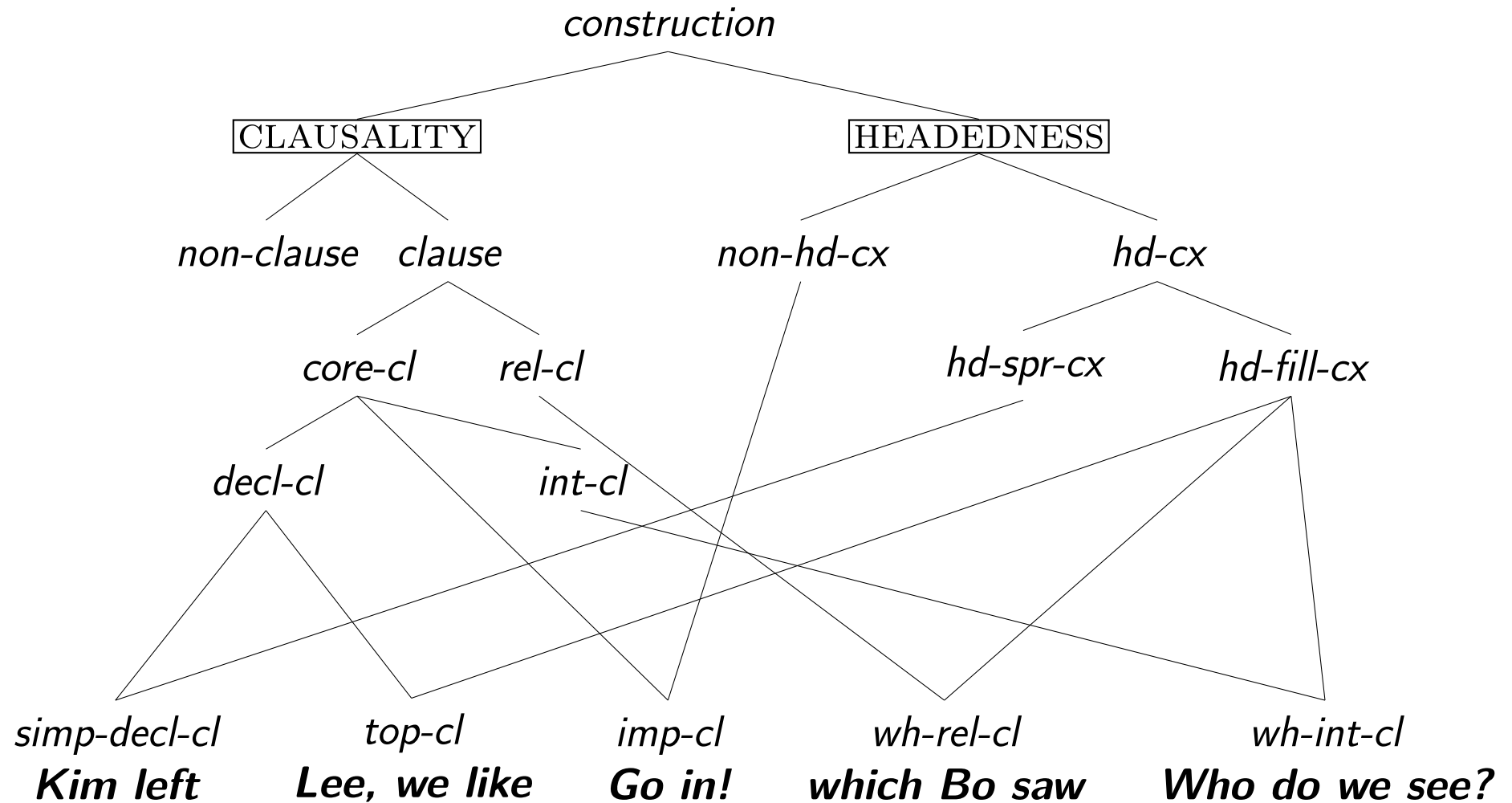
Overview of Differences (SBCG vs HPSG)

- Multiple Inheritance
- Signs
- Grammar rules form a hierarchy — many more rules
- Every tree node has its own phonology
- Many principles become constraints on grammar rules
- The definition of well-formedness is simplified

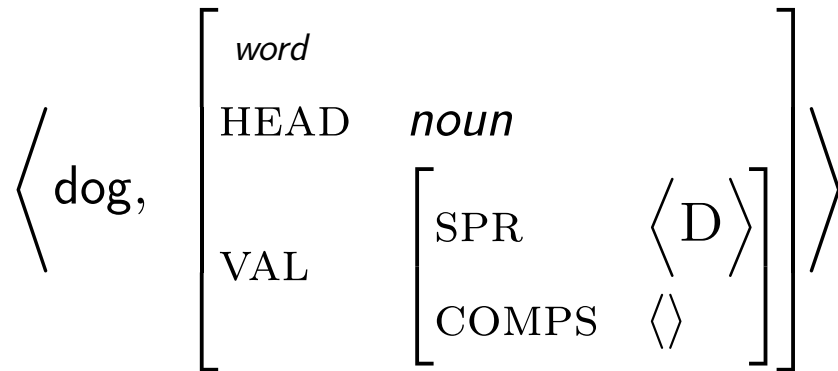
More Inheritance (lexemes)



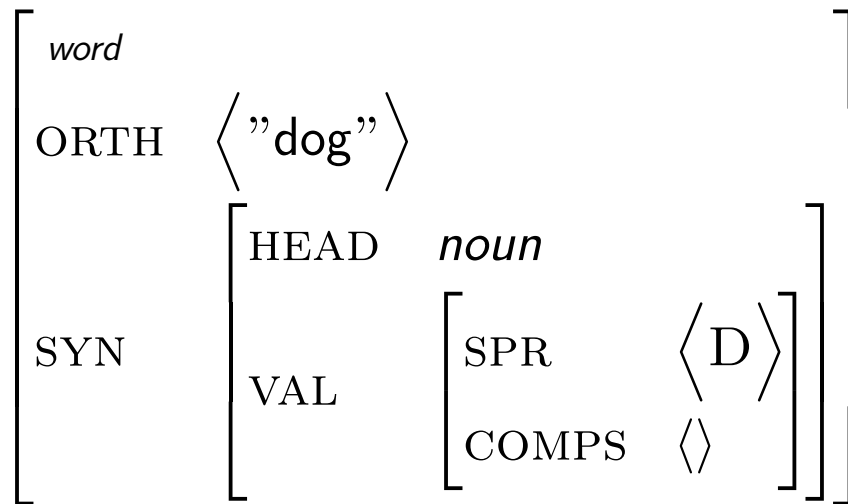
More Inheritance (clauses)



Words are signs



Becomes:



Rules are signs

$$\left[\begin{array}{l} \textit{phrase} \\ \text{VAL} \left[\begin{array}{l} \text{COMPS} \langle \rangle \\ \text{SPR} \langle \rangle \end{array} \right] \end{array} \right] \rightarrow \boxed{2} \textbf{H} \left[\text{VAL} \left[\begin{array}{l} \text{COMPS} \langle \rangle \\ \text{SPR} \langle \boxed{2} \rangle \end{array} \right] \right]$$

Becomes:

$$\left[\begin{array}{l} \textit{head-spr-cx} \\ \text{MOTHER} \left[\text{VAL.SPR} \langle \rangle \right] \\ \text{HD-DTR} \boxed{2} \left[\text{VAL} \left[\begin{array}{l} \text{SPR} \langle \boxed{1} \rangle \\ \text{COMPS} \langle \rangle \end{array} \right] \right] \\ \text{DTRS} \langle \boxed{1}, \boxed{2} \rangle \end{array} \right]$$

Principles are signs

- Head Feature Principle

$$\begin{bmatrix} \text{MOTHER.SYN.HEAD} & \boxed{1} \\ \text{HD-DTR.SYN.HEAD} & \boxed{1} \end{bmatrix}$$

- Semantic Inheritance Principle

$$\begin{bmatrix} \text{MOTHER.SEM} & \begin{bmatrix} \text{MODE} & \boxed{1} \\ \text{INDEX} & \boxed{2} \end{bmatrix} \\ \text{HD-DTR.SEM} & \begin{bmatrix} \text{MODE} & \boxed{1} \\ \text{INDEX} & \boxed{2} \end{bmatrix} \end{bmatrix}$$

- Valence Principle

$$\begin{bmatrix} \text{MOTHER.SYN.VAL} & / \boxed{1} \\ \text{HD-DTR.SYN.VAL} & / \boxed{1} \end{bmatrix}$$

The Root Condition (Start Symbol) is a sign

$$\left[\begin{array}{c} \text{SYN} \\ \text{GAP} \end{array} \left[\begin{array}{c} \text{HEAD} \\ \text{VAL} \end{array} \left[\begin{array}{c} \left[\begin{array}{c} \text{verb} \\ \text{FORM} \end{array} \right] \text{fin} \\ \left[\begin{array}{c} \text{COMPS} \\ \text{SPR} \end{array} \right] \langle \rangle \end{array} \right] \right] \right]$$

Sign-based Construction Grammar

- SBCG makes the analyses more uniform
- Puts information just where it is needed
- Merges the insights of HPSG and Construction Grammar
- Most recent work on HPSG is effectively work on SBCG

Look at Ian Sag's slides if you want to know more!